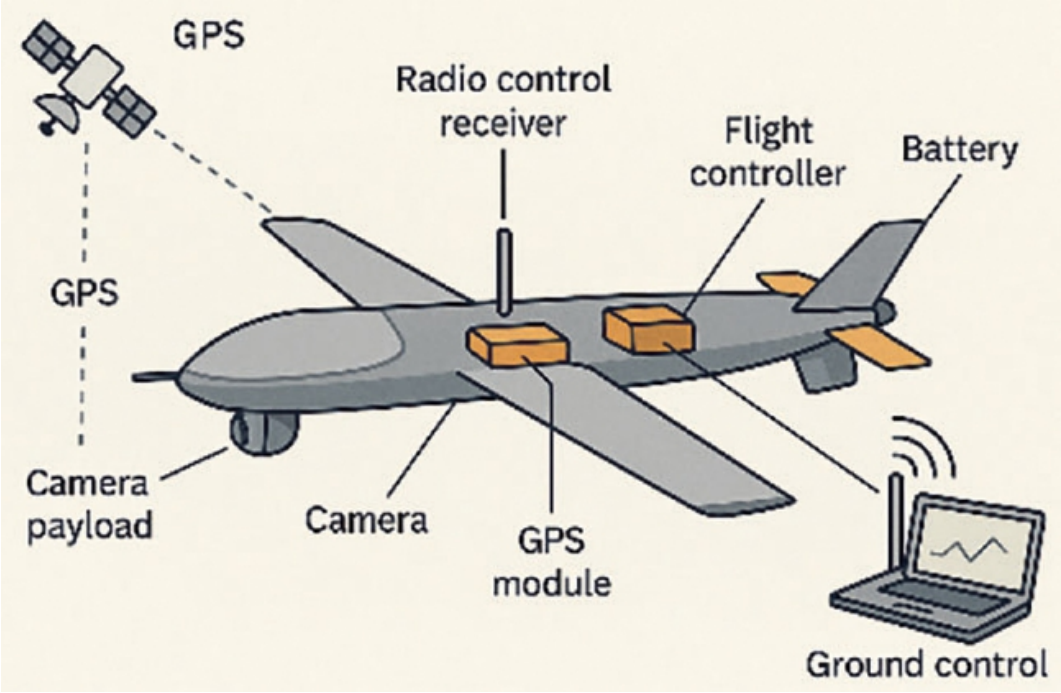


Figure 1.4 Second-generation drone remote control and GPS integration. ⏎

1.4.3 Third generation: Surveillance and combat capabilities

The third generation of unmanned aerial systems (UAS) represents a decisive shift from utility and navigation improvements toward operational maturity and mission-centric design, with platforms engineered specifically for sustained intelligence, surveillance, reconnaissance (ISR), and strike roles. Emerging in earnest from the late 1990s through the 2010s, this generation is characterized by long-endurance airframes, higher-altitude operations, robust payload integration, secure datalinks, and hardened avionics that enable persistent, command-driven missions in contested environments.

Technically, third-generation drones combine several key subsystems optimized for operational effectiveness. High-efficiency propulsion and larger fuel/power reserves provide multi-hour to multi-day endurance; stabilized gimbal sensors (EO/IR, synthetic aperture radar) allow precision imaging and moving-target indication; multi-band datalinks and anti-